

Pandas — LEVEL 11: Basic Statistics

Line-by-line code, output, examples, and easy interview explanations.

1. sum()

Code

```
df = pd.DataFrame({"Score": [70, 80, 90, 100]})  
print(df["Score"].sum())
```

Output

```
340
```

Explanation: sum() adds all values.

Interview: Used for totals such as total sales.

2. mean()

Code

```
print(df["Score"].mean())
```

Output

```
85.0
```

Explanation: mean() calculates the average.

Interview: Mean = total ÷ number of values.

3. median()

Code

```
print(df["Score"].median())
```

Output

```
85.0
```

Explanation: median() returns the middle value after sorting.

Interview: Median is less affected by extreme values.

4. mode()

Code

```
s = pd.Series([70, 80, 80, 90])  
print(s.mode())
```

Output

```
0      80  
dtype: int64
```

Explanation: mode() returns the most frequent value.

Interview: A dataset can have more than one mode.

5. min()

Code

```
print(df["Score"].min())
```

Output

```
70
```

Explanation: min() returns the smallest value.

Interview: Useful for finding the lowest value.

6. max()

Code

```
print(df["Score"].max())
```

Output

```
100
```

Explanation: max() returns the largest value.

Interview: Useful for finding the highest value.

7. std()

Code

```
print(round(df["Score"].std(), 2))
```

Output

```
12.91
```

Explanation: std() measures how spread out values are around the mean.

Interview: Higher standard deviation means more spread.

8. var()

Code

```
print(round(df["Score"].var(), 2))
```

Output

```
166.67
```

Explanation: var() measures variance, the square of standard deviation.

Interview: Pandas uses sample variance by default (ddof=1).

9. count()

Code

```
df = pd.DataFrame({"Score": [70, 80, None, 100]})  
print(df["Score"].count())
```

Output

```
3
```

Explanation: count() counts non-missing values.

Interview: Unlike `count()`, missing values are not counted.

10. quantile()

Code

```
df = pd.DataFrame({"Score": [70, 80, 90, 100]})
print(df["Score"].quantile(0.50))
```

Output

```
85.0
```

Explanation: `quantile()` returns a value at a chosen percentile.

Interview: 0.50 is the 50th percentile, equivalent to the median.

11. describe()

Code

```
df = pd.DataFrame({"Score": [70, 80, 90, 100]})
print(df["Score"].describe())
```

Output

```
count      4.000000
mean       85.000000
std        12.909944
min        70.000000
25%        77.500000
50%        85.000000
75%        92.500000
max        100.000000
Name: Score, dtype: float64
```

Explanation: `describe()` gives count, mean, std, min, quartiles, and max.

Interview: It is a quick statistical summary of a numeric column.

12. cumsum()

Code

```
df = pd.DataFrame({"Sales": [100, 200, 300, 400]})
print(df["Sales"].cumsum())
```

Output

```
0      100
1      300
2      600
3     1000
Name: Sales, dtype: int64
```

Explanation: `cumsum()` creates a running total.

Interview: Useful for cumulative sales or running totals.

13. cumprod()

Code

```
df = pd.DataFrame({"Value": [2, 3, 4, 5]})
print(df["Value"].cumprod())
```

Output

```
0      2
1      6
2     24
3    120
Name: Value, dtype: int64
```

Explanation: cumprod() creates a running product.

Interview: Each result multiplies all values up to the current row.

14. cummin()

Code

```
df = pd.DataFrame({"Sales": [500, 300, 400, 200]})
print(df["Sales"].cummin())
```

Output

```
0      500
1      300
2      300
3      200
Name: Sales, dtype: int64
```

Explanation: cummin() returns the smallest value seen so far.

Interview: Useful for tracking a running minimum.

15. cummax()

Code

```
df = pd.DataFrame({"Sales": [500, 700, 600, 900]})
print(df["Sales"].cummax())
```

Output

```
0      500
1      700
2      700
3      900
Name: Sales, dtype: int64
```

Explanation: cummax() returns the largest value seen so far.

Interview: Useful for tracking a running maximum.

QUICK REVISION

sum()	-> Total
mean()	-> Average
median()	-> Middle value
mode()	-> Most frequent value
min()	-> Smallest
max()	-> Largest
std()	-> Standard deviation
var()	-> Variance
count()	-> Non-missing count
quantile()	-> Percentile value
describe()	-> Statistical summary
cumsum()	-> Running total
cumprod()	-> Running product
cummin()	-> Running minimum
cummax()	-> Running maximum